MS&E 226 — Midterm 1 Practice Questions (Generated Set 1): Solutions

Question 1

A learning algorithm predicts an outcome at a fixed covariate vector (x). At this point, the irreducible error is $\sigma^2((x)) = 9$, the bias is -2, and the variance is 5. What is the resulting Average MSE?

(a) 18

(b) 12

(c) 9

(d) $\sqrt{18} \approx 4.24$

(e) I choose not to answer this question.

Answer: (a)

Average MSE = irreducible error + Bias^2 + Variance $= 9 + (-2)^2 + 5 = 9 + 4 + 5 = 18$.

(a) Correct — the arithmetic above.

(b) Uses Bias instead of Bias^2: $9 + (-2) + 5 = 12$. The decomposition requires squaring the bias; a negative bias still costs you positive MSE.

(c) Drops the irreducible error term entirely ($(-2)^2 + 5 = 9$) — this is the **estimator** MSE formula from Lecture 8 (MSE $=$ Bias^2 $+$ Variance, no irreducible error), not the **prediction** MSE formula from Lecture 7. Mixing up the two decompositions is the single easiest way to get this kind of question wrong.

(d) Confuses MSE with RMSE by taking a square root that doesn't belong here.

Question 2

A fellow student claims: "There's always a strict bias-variance tradeoff: whenever you change your learning algorithm so that variance decreases, bias must increase, and vice versa." Which of the following best evaluates this claim, based on what was covered in lecture?

(a) Correct — this follows directly from the bias-variance decomposition, since Average MSE = irreducible error + Bias^2 + Variance, and the three terms must always sum to a fixed total.

(b) Correct — the Gauss-Markov theorem shows OLS has the lowest variance among all zero-bias estimators using a given feature set, so no learning algorithm can ever have both lower bias and lower variance than another.

(c) Incorrect — the decomposition is just that, a decomposition, not a conservation law. Nothing forces bias and variance to move in opposite directions, and lecture gave explicit examples (double descent; adding genuinely informative features) where both bias and variance decreased at the same time.

(d) Incorrect — in fact the opposite is true: bias and variance always move together in the **same** direction as model complexity increases.

(e) I choose not to answer this question.

Answer: (c)

(a) Misreads "decomposition" as "fixed total." Nothing constrains Average MSE to stay constant — a learning algorithm can simply get better (lower MSE) by improving bias and variance simultaneously.

(b) The Gauss-Markov theorem is real, but it only applies to a specific comparison: linear, **unbiased** estimators using a **fixed** feature set. It says nothing about comparisons across different feature sets or non-linear algorithms, which is exactly the loophole double descent and the weight-measurement example exploit.

(c) Correct. The bias-variance decomposition, Average MSE = irreducible error + Bias^2 + Variance, is exactly that — a decomposition of whatever MSE happens to be, not a conservation law that forces the pieces to trade off. Lecture gave concrete counterexamples: the repeated weight-measurement example (both bias and variance fell as more measurements were added) and the double-descent lasso example (adding genuinely informative features reduced both bias and variance, and even reduced the "irreducible" error relative to the enlarged feature set).

(d) Overcorrects to the opposite blanket claim. Bias and variance don't always move together either (e.g., regularization's whole point is moving them in opposite directions) — there's simply no universal law either way.

Question 3

Assume (A1)–(A3) hold (linear population model; all relevant features observed; homoskedastic, mean-zero errors). A data scientist fits a ridge regression model with $\lambda > 0$ on the same features used by an OLS model, and finds the ridge model has strictly lower variance than OLS at every test point. What must be true of the ridge model's bias, and why?

(a) It must be exactly zero too, since ridge regression — like OLS — is a linear, unbiased estimator under (A1)–(A3).

(b) It must be nonzero (strictly positive, in the sense of having nonzero bias at some point), because the Gauss-Markov theorem says OLS has the lowest variance among **all** linear, unbiased estimators using these features — so a linear estimator with strictly lower variance than OLS cannot also be unbiased.

(c) It could be anything — positive, negative, or zero — since bias and variance are unrelated quantities under (A1)–(A3).

(d) It must be **lower** than OLS's bias (i.e., closer to zero), since regularization is specifically designed to correct bias in the OLS estimate.

(e) I choose not to answer this question.

Answer: (b)

(a) Contradicts the Gauss-Markov theorem directly: no unbiased linear estimator can beat OLS's variance.

(b) Correct. Under (A1)–(A3), the Gauss-Markov theorem says OLS has the **lowest variance among all linear, unbiased estimators** that use the same features. If the ridge estimator (which is linear in $\boldsymbol{Y}$) has strictly lower variance than OLS, it cannot be unbiased — otherwise it would contradict OLS's optimality within that class. So the ridge model must have nonzero bias somewhere.

(c) Ignores Gauss-Markov. This is the right instinct when comparing across **different feature sets** or **different algorithm families** (Question 2's lesson), but Gauss-Markov gives a precise, forced relationship for **this specific** comparison (ridge vs. OLS, same features, under A1–A3) — context matters.

(d) Backwards. Regularization's mechanism is to **add** bias in exchange for variance reduction, not to "correct" an existing OLS bias (OLS already has zero bias under A1–A3, so there's no bias to correct).

Question 4

Assume (A1)–(A3) hold for a population model with three relevant features A, B, C: $Y = \beta_A A + \beta_B B + \beta_C C + \varepsilon$. A data scientist fits OLS using only A and B, omitting C. Which of the following is the most accurate statement about the resulting model?

(a) The model is unbiased — OLS remains BLUE (Best Linear Unbiased Estimator) even when a relevant feature is omitted, as long as the remaining features are observed correctly.

(b) The model is generally biased, and the size of that bias does not depend at all on how C is correlated with A and B — omitting any relevant feature produces the same bias regardless of correlation structure.

(c) The model is generally biased, and the size of the bias — along with any distortion to the fitted coefficients on A and B themselves — depends on how correlated the omitted feature C is with A and B.

(d) The model is biased, but only in its predictions — the fitted coefficients on A and B remain correct (unbiased) estimates of β_A and β_B.

(e) I choose not to answer this question.

Answer: (c)

(a) Gauss-Markov's zero-bias guarantee requires (A2) — **all** relevant features observed. Omitting C violates this, so the guarantee simply doesn't apply here.

(b) Directly contradicts the lecture's point that the bias "depends on how correlated the remaining features are with the omitted features."

(c) Correct, per the lecture's remarks on omitted-variable bias (OVB): omitting a relevant feature breaks assumption (A2) (not all relevant features observed), so the Gauss-Markov zero-bias guarantee no longer applies, and both the size of the resulting bias and the distortion to the fitted coefficients on the **included** features depend on the correlation between the omitted and included features.

(d) Contradicts the omitted-variable-bias (OVB) phenomenon explicitly named in lecture: omitting a correlated feature can corrupt the **coefficients** on the included features too, not just the final predictions.

Question 5

In Lecture 7, the bias-variance decomposition for a prediction $\hat{f}((x))$ includes an irreducible error term $\sigma^2((x))$. In Lecture 8, the analogous decomposition for an estimator $\hat{\theta}$ of a population parameter θ is simply $\text{MSE} = \text{Bias}^2 + \text{Variance}$, with **no** irreducible error term. Why does the estimator decomposition have no irreducible error term?

(a) Because estimators are always unbiased, so there is nothing left to attribute to irreducible error.

(b) Because the irreducible error term only matters for classification problems, not for estimating a continuous parameter.

(c) Because the true outcome Y being predicted in the prediction problem is itself random even conditional on (x), whereas the parameter θ being estimated is treated as a **fixed** (non-random) quantity under the frequentist view — so there's no analogous "irreducible" randomness in the target itself.

(d) Because irreducible error only arises when the sample size is small; for the large-n asymptotic theory used in estimation, it vanishes.

(e) I choose not to answer this question.

Answer: (c)

(a) False in general — plenty of estimators are biased (e.g., OLS with an omitted feature, from Question 4). Being unbiased isn't why the irreducible-error term disappears; the reason is about what kind of thing the **target** is (random outcome vs. fixed parameter), not about the estimator's own bias.

(b) Invents a restriction that isn't in the material — irreducible error is a general prediction-theory concept, not classification-specific.

(c) Correct. Irreducible error in the prediction setting exists because the **target** Y is itself a random variable with leftover variance even given (x) exactly (i.e., $\sigma^2((x)) = \text{Var}(Y \mid (x)) > 0$ in general). In the estimation setting, the target is a population parameter θ, which the frequentist view treats as a **fixed**, non-random quantity — there's no analogous leftover randomness in a fixed number, so there's nothing for an "irreducible error" term to capture.

(d) Confuses two different "shrinks with n" facts: it's the **variance** of an estimator that shrinks as n grows, not irreducible error (which, in the prediction setting, is about the conditional variance of Y given (X) and has nothing to do with how much data you have).

Question 6

Under the frequentist view of statistical inference, which of the following is randomly varying across "parallel universes," and which is fixed?

(a) The population parameter θ is random; the data sample is fixed.

(b) Both the population parameter θ and the data sample are random.

(c) The population parameter θ is fixed; the data sample (and therefore any estimator computed from it) is random.

(d) Both the population parameter θ and the data sample are fixed once the sample has been collected — there is no remaining randomness to reason about.

(e) I choose not to answer this question.

Answer: (c)

(a) Reverses the frequentist view (closer to how some people informally imagine a Bayesian perspective, but even that isn't quite this).

(b) Treats the parameter as random too, which the frequentist view explicitly rejects.

(c) Correct — this is the core frequentist premise: the true parameter θ is fixed and unknown; the only randomness is in which particular data sample you happened to draw (the "parallel universes" thought experiment), and any estimator computed from that sample inherits its randomness.

(d) A common but mistaken intuition: once you're holding a specific, already-collected dataset, the **numbers in front of you** are fixed, but that doesn't mean there was no randomness in the process that produced them. The parallel-universes framework reasons about the data-generating process, not about a single realized dataset in isolation.

Question 7

You estimate a population mean μ using the sample mean of n i.i.d. draws. With a sample of size $n = 25$ you compute standard error $\widehat{\mathrm{SE}}_{25}$; with a much larger sample of size $n = 400$ (and roughly the same sample standard deviation $\hat{\sigma}$) you compute $\widehat{\mathrm{SE}}_{400}$. Approximately how do these compare?

(a) $\widehat{\mathrm{SE}}_{25}$ is about 4 times larger than $\widehat{\mathrm{SE}}_{400}$.

(b) $\widehat{\mathrm{SE}}_{25}$ is about 16 times larger than $\widehat{\mathrm{SE}}_{400}$.

(c) $\widehat{\text{SE}}_{25}$ and $\widehat{\text{SE}}_{400}$ are approximately equal, since standard error doesn't depend on sample size once $\hat{\sigma}$ is fixed.

(d) $\widehat{\text{SE}}_{400}$ is about 4 times larger than $\widehat{\text{SE}}_{25}$.

(e) I choose not to answer this question.

Answer: (a)

$\text{SE} = \frac{\hat{\sigma}}{\sqrt{n}}$, so the ratio of standard errors is the **inverse square root** of the ratio of sample sizes:

$$\frac{\widehat{\text{SE}}_{25}}{\widehat{\text{SE}}_{400}} = \frac{\sqrt{400}}{\sqrt{25}} = \frac{20}{5} = 4.$$

So $\widehat{\text{SE}}_{25}$ is about 4 times larger than $\widehat{\text{SE}}_{400}$.

(a) Correct, by the computation above.

(b) Uses the ratio of sample sizes directly ($\frac{400}{25} = 16$) instead of the ratio of their square roots — forgetting that SE scales with $\frac{1}{\sqrt{n}}$, not $\frac{1}{n}$.

(c) Confuses the standard deviation of individual observations (which doesn't shrink with n) with the standard error of the sample **mean** (which does, proportionally to $\frac{1}{\sqrt{n}}$) — precisely the distinction lecture called out as important to keep straight.

(d) Gets the direction backwards: **more** data means a **smaller** standard error, not a larger one, so it's $\widehat{\text{SE}}_{25}$ that's bigger.

Question 8

After collecting a sample and computing a 95% confidence interval $[28.1, 35.7]$ for a population mean μ, a colleague says: "There is a 95% probability that μ is between 28.1 and 35.7." Is this an accurate interpretation, as discussed in lecture?

(a) Yes — by definition, a 95% confidence interval means there's a 95% probability the true parameter falls inside this specific interval.

(b) No — the 95% figure describes the **procedure**: across repeated samples, 95% of the resulting intervals would contain the true, fixed μ. Once a specific interval like $[28.1, 35.7]$ has been computed, μ either is or isn't in it; μ is not a random variable, so it doesn't make sense to assign it a probability.

(c) No — the correct statement is that there's a 95% probability the **sample mean** $\overline{Y}$ falls between 28.1 and 35.7.

(d) Yes, but only if the underlying population distribution is itself normal; otherwise the 95% figure doesn't apply.

(e) I choose not to answer this question.

Answer: (b)

(a) This is exactly the tempting-but-incorrect interpretation flagged in lecture: it treats μ as if it were still random after the interval has been computed.

(b) Correct. A confidence interval's coverage is a statement about the **procedure**, evaluated across repeated samples (parallel universes): 95% of the intervals this procedure produces will contain the true μ. Once you've plugged in one actual sample and computed one specific interval, there's no more randomness left in μ to assign a probability to — μ is fixed; the interval (built from your random sample) was the random object, before you computed it.

(c) Misapplies the 95% figure to $\overline{Y}$ instead of μ — and doesn't even make sense on its own terms, since a symmetric CI is centered exactly on the observed $\overline{Y}$, which trivially lies at the interval's midpoint, not in it "with 95% probability."

(d) Not required: the Central Limit Theorem guarantees the sampling distribution of $\overline{Y}$ is approximately normal for large n **regardless of the shape of the individual Y_i's distribution** — that was explicitly called out in lecture as the "remarkable" part of the CLT.

Question 9

Holding the sample size n and sample standard deviation $\hat{\sigma}$ fixed, consider two confidence intervals for the population mean constructed from the same sample: a 95% CI and a 99% CI. Evaluate:

(i) The 99% CI is wider than the 95% CI.

(ii) The 99% CI is guaranteed to contain the true population mean μ, while the 95% CI is not.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements are correct.

(d) Neither statement is correct.

(e) I choose not to answer this question.

Answer: (a)

(i) is correct. A higher confidence level requires a larger critical value ($z_{\alpha/2} \approx 2.58$ for 99% vs. 1.96 for 95%), which directly widens the interval $\overline{Y} \pm z_{\alpha/2}\widehat{\text{SE}}$, holding n and $\hat{\sigma}$ (and hence $\widehat{\text{SE}}$) fixed.

(ii) is incorrect. "Guaranteed to contain μ" describes neither interval. Both are built from the same random sample and both have only a long-run **coverage rate** (99% or 95%, respectively, across repeated

sampling) — not a guarantee for this one specific interval. This is the same misconception as Question 8, applied to a comparison between two intervals instead of one.

Since (i) is correct and (ii) is not, the answer is (a).

(a) Correct — see the reasoning above.

(b) (ii) alone is not correct; see above.

(c) (ii) is not correct, so both cannot be.

(d) (i) is correct, so it isn't true that neither is.

Question 10

Consider the sample mean $\overline{Y}$ computed from n i.i.d. draws from a population with mean μ and variance σ^2. The same quantity $\overline{Y}$ can be analyzed two ways covered in this course: (I) as a **prediction** of a new, as-yet-unseen outcome Y from the same population, and (II) as an **estimator** of the population parameter μ itself. Which of the following correctly compares the Bias and Variance terms, and the overall MSE, across these two analyses?

(a) Bias and Variance are different in the two analyses, because predicting Y requires a different formula for $\overline{Y}$ than estimating μ.

(b) Bias and Variance are identical in the two analyses — in both cases Bias $= 0$ and Variance $= \frac{\sigma^2}{n}$ — but the overall MSE differs, because the prediction-MSE additionally includes the irreducible error term σ^2, which has no counterpart when the target is the fixed parameter μ.

(c) Bias and Variance are identical, and so is the overall MSE in both analyses, since $\overline{Y}$ is being compared to the same kind of target (μ) in both cases.

(d) Only the Variance term is identical across the two analyses ($\frac{\sigma^2}{n}$ in both cases); the Bias differs, because as a predictor $\overline{Y}$ is biased (due to the irreducible error), while as an estimator it is exactly unbiased.

(e) I choose not to answer this question.

Answer: (b)

$\overline{Y}$ is literally the same number, computed the same way, in both analyses — so its Bias and Variance (as derived in Lecture 7 for the sample mean specifically) are identical regardless of which question you're using it to answer: Bias $= 0$ and Variance $= \sigma^2/n$. What differs is only the **target** being compared against:

- As a **prediction** of a new outcome Y (Lecture 7): Y itself is random even once you know μ exactly, contributing an irreducible error term σ^2. Average MSE $= \sigma^2 + 0 + \sigma^2/n$.
- As an **estimator** of μ (Lecture 8): μ is a fixed parameter under the frequentist view, so there's no analogous leftover randomness to contribute an irreducible term. MSE $= 0 + \sigma^2/n$.

(a) False — $\overline{Y}$ doesn't change formula between the two uses; it's the exact same statistic playing two different roles.

(b) Correct, matching the reasoning above.

(c) Ignores the irreducible error term entirely; the two MSEs differ by exactly σ^2 (see Question 5's reasoning for why that term appears in one case and not the other).

(d) Garbles which term is "the same" (it's Variance **and** Bias that match, not just Variance) and misattributes irreducible error to Bias — irreducible error is a separate, third term in the decomposition, not a form of bias.